SECTION A

*Write about a quarter of a page each on any **four** of the following topics.*

Deduct marks for each major thing missed, and for each wrong statement. In general, be relatively generous if the answer makes sense and contains the main ideas.

3. *The MAPE is better than the RMSE for measuring forecast accuracy because it is easier to explain.*

   - The MAPE is a percentage, so it is easier to interpret than the RMSE. 1
   - However, the MAPE may not make sense for the data, especially if the data contains zeros, small values, or negative values, or if the zero point is arbitrary. 1
   - The RMSE is scale-dependent so can't be used to compare forecasts across different data sets, whereas the MAPE is scale-independent. 1
   - The RMSE is optimized by the mean, so finding a model that minimizes the RMSE will give the best point forecasts. 1
   - The MAPE is not optimal for any meaningful characteristic of the forecasts. 1

6. *STL is not useful for forecasting because it only provides a decomposition of the data.*

   - STL is a decomposition method that separates a time series into trend, seasonal, and remainder components. 1
   - STL, on its own, is not a forecasting method. 1
   - However, STL is useful for forecasting because the decomposition can help us understand the data better. 1
   - STL can be coupled with a forecasting method applied to the seasonally adjusted data. 1
   - In this case, the seasonal component is usually forecast using the seasonal naive method, and then the seasonal component forecasts and seasonally adjusted forecasts can be added to obtain forecasts of the original data. 1

SECTION B

Figures 1–3 relate to the daily use of public transport in Canberra, from July 2019 – March 2024. The variable plotted is the total number of passenger boardings each day on all forms of public transport except for school buses.

3. *You have been asked to provide forecasts for the next four weeks for daily passenger boardings. Consider applying each of the methods and models below to the data from March 2022 onwards. Comment, in a few words each, on whether each one is appropriate for forecasting the next two weeks of data, and what features of the data each would miss. No marks will be given for simply guessing whether a method or a model is appropriate without justifying your choice.*

Be flexible in marking, especially around the challenge with capturing term breaks and holidays.

*Start your response by stating: **suitable** or **not suitable**.*

(a) *Seasonal naïve method using weekly seasonality.*

There are always grey areas

   - Maybe suitable, but will miss the annual seasonality and holidays. 1

(b) *Naïve method*

   - Unsuitable. Won't even capture weekly seasonality. 1